



FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3123 - BUSINESS INTELLIGENCE

SECTION 2

LECTURER: DR. NOORFA HASZLINNA BT MUSTAFFA

PROJECT PROPOSAL

TITLE:

**BUSINESS INTELLIGENCE ANALYTICS FOR U.S. E-COMMERCE PERFORMANCE
OPTIMIZATION: A MULTI-DIMENSIONAL ANALYSIS OF SALES, CUSTOMER
BEHAVIOUR, AND DELIVERY OPERATIONS (2019–2025)**

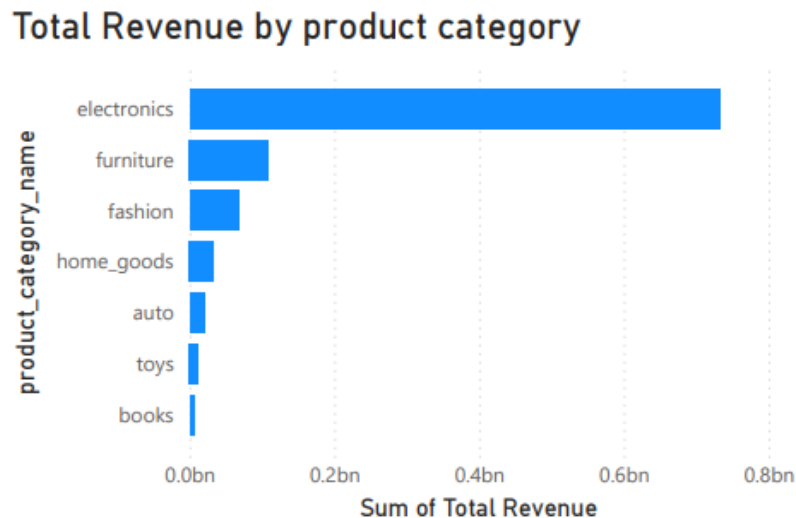
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1.0 Revenue & Product Analysis (Muhammad Syahmi Faris bin Rusli)

This module examines the company's revenue performance across product categories and over time, identifying which segments drive the most business value.

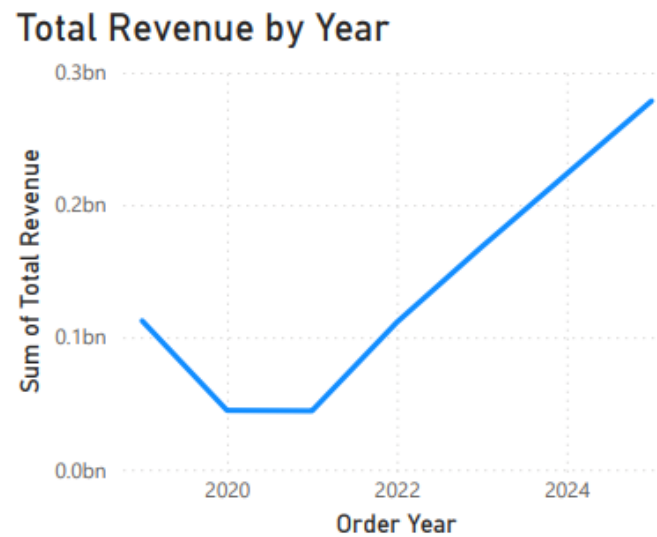
1.1 Total Revenue by Product Category



Business Insight: **Electronics** is the dominant revenue category, generating approximately **732.78 million**, far exceeding all other categories. **Furniture** is **109.09 million** and **fashion 67.82 million** rank second and third, while home goods, auto, toys, and books contribute smaller shares. The heavy concentration in electronics presents a revenue risk should demand in that segment decline.

Business Recommendation: The company should strengthen its electronics supply chain to protect its primary revenue stream. Simultaneously, targeted promotions and cross-selling strategies such as bundling electronics with furniture or home goods which can help grow underperforming categories and reduce revenue concentration risk.

1.2 Revenue Trend Over Time



Business Insight: Revenue grew steadily from 2020 through 2022-2023, driven by post pandemic e-commerce expansion. Growth has since begun to plateau, suggesting the business is approaching organic market saturation within its existing footprint.

Business Recommendation: To re-accelerate growth, the company should introduce a customer loyalty programme to improve retention and repeat purchase rates. Geographic expansion beyond the current 10-state footprint and personalised marketing based on existing order data can help unlock new revenue opportunities.

2.0 Logistics and Delivery Analytics (Afif Shaqir Irfan bin Arqam)

This module examines the company's regional logistic performance by identifying which part of US states take the higher order volume and the extent to which delivery delays vary across the network.

2.1 Order Volume and Delivery Delay Rate by State

seller_state	Count of order_id	Sum of is_delayed
AZ	73895	7296
CA	447809	44125
CO	86179	8476
FL	258274	25334
GA	73912	7375
IL	226492	22182
NY	369051	36189
PA	74287	7317
TX	284394	27829
WA	159794	15818
Total	2054087	201941

Business Insight : The regional shows that **California (CA)** handles the highest order volume with **447,809 orders (21.79% of total volume)**, followed by **New York (NY)** with **369,051 orders (17.97%)** and **Texas (TX)** with **284,394 orders (13.85%)**. Together, these three states account for over half of the company's total order volume, confirming their role as the primary operational hubs in the network. These same high-volume states also record the highest absolute number of delayed orders and California alone accounts for **44,125 delayed orders**, followed by New York (**36,189**) and Texas (**27,829**). However, when delay counts are converted into a delay rate (delayed orders ÷ total orders per state), a different pattern emerges: every state falls within a narrow band of **9.79% to 9.98%**, with **Georgia** recording the highest rate at **9.98%** and **Texas and Illinois** the lowest at **9.79%**. Across the entire network, **201,941 of 2,054,087 orders were delayed**, an overall delay rate of **9.83%**. The narrow spread between the highest and lowest state-level delay rates indicates that delivery delays are not concentrated in any particular region,

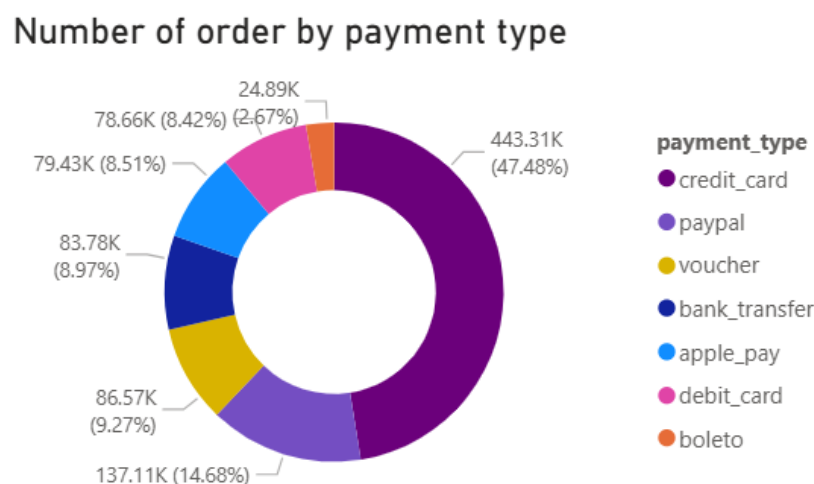
but instead reflect a systemic issue affecting roughly one in every ten orders regardless of where the order is fulfilled.

Business Recommendation: Because California, New York, and Texas drive the majority of order volume, the company should prioritise warehouse capacity, carrier contracts, and staffing in these states to prevent operational strain as volume continues to grow.

3.0 Customer & Payment Analytics (Choh Jing Yi)

This module analyzes consumer financial behavior, highlighting how customers choose to pay and how those choices correlate with overall satisfaction.

3.1 Payment Method Distribution

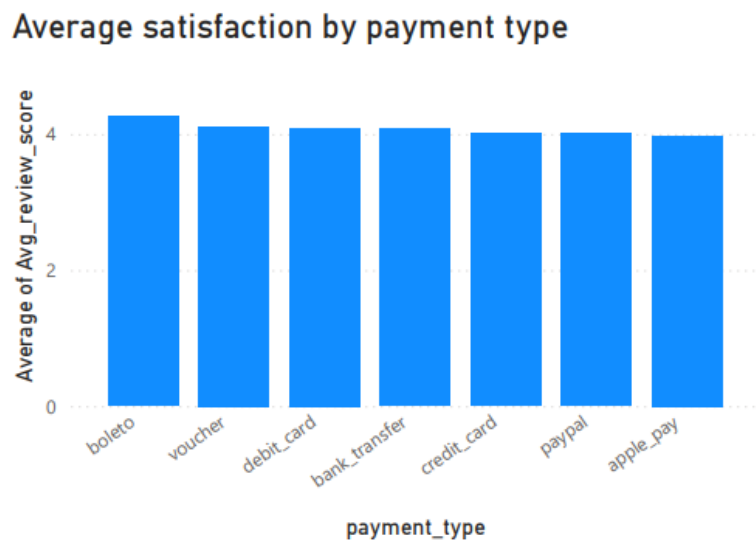


Business Insight: The payment method distribution shows that **credit card** is the dominant payment option in nearly half of all transactions about **443.31K orders (47.48%)**. This indicates that customers strongly prefer credit cards due to their convenience, familiarity and widespread acceptance. **PayPal** ranks as the second most popular payment method with **137.11K orders (14.68%)** highlighting a significant customer segment that prefers digital wallet solutions for faster and more secure online payments. The remaining payment methods include **voucher (9.27%)**, **bank transfer (8.97%)**, **Apple Pay (8.51%)**, and **debit card (8.42%)**, contributing

relatively similar shares suggesting that customers value having multiple payment options available. Meanwhile, **boleto** records the lowest usage at **24,889 orders (2.67%)** indicating limited customer preference for this payment method. The overall distribution suggests that credit cards are still the dominant payment method for people. Alternative digital payment methods make up a significant proportion of payments, representing different people's payment options.

Business Recommendation: The majority of transactions mainly focus on credit card payments. So, IT operations must prioritize the stability and security of the credit card payment gateway to ensure the reliability, security, and availability of the credit card payment gateway. If there is any disruption to this service, it may have a significant effect on revenue generation. In addition, the company can consider promotional initiatives such as cashback offers or discounts to encourage customers to use lower cost payment methods like Bank Transfers or Debit Cards to reduce transaction processing expenses.

3.2 Average Review Score by Payment Type



Business Insight: Customer satisfaction remains consistently high across all payment methods, with average review scores ranging from **3.98 to 4.25 out of 5**. This indicates that customers generally have a positive payment experience regardless of the method they choose. Among all payment options, **Boleto** records the highest average review score of **4.25**, suggesting that users

who prefer this payment method are particularly satisfied with their purchasing experience. On the other hand, **Apple Pay** receives the lowest average score of **3.98**. The difference compared to other methods is much smaller.

The narrow gap between payment methods suggests that the payment system is reliable and does not create significant customer dissatisfaction. Overall, the findings reflect a well performing payment infrastructure that supports a positive customer experience.

Business Recommendation: The company has consistently high customer satisfaction with all payment methods, so they should maintain the stability and reliability of their payment systems. The high satisfaction score of Boleto provides an opportunity to understand what aspects of the payment experience customers appreciate and determine whether similar features can be applied to other payment methods.

Apple Pay's score is still positive, 3.98 out of 5. It is just slightly lower than other payment methods. The business should investigate the Apple Pay payment process especially on the mobile app to understand any user experience problems, delays or technical issues that could impact the customer experience. Regular reviews to feedback related to payment allows for continued adaptation of payment methods to satisfy customer expectations and support a smooth checkout process.